

Effective Independence: Quantifying the Reliability Ceiling of Same-Base-Model LLM Ensembles

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Abstract—Debate, self-consistency, and LLM-as-judge pipelines assume ensemble diversity they do not actually have when every voter is sampled from the same base model. We apply the classical effective-sample-size statistic N_{eff} to this setting and measure it across four diversity axes commonly used to build LLM ensembles (prompt, temperature, model size, and model family) on three reasoning and factuality benchmarks: GPQA-Diamond, MATH levels 4-5, and SimpleQA. Measured with the usual correlation coefficient, even $N = 3$ ensembles behave like barely more than a single independent voter ($N_{\text{eff}} = 1.08$ for prompt diversity, 1.43 for model-family diversity, 1.37 for all 12 configurations together). We then show that this coefficient flatters such ensembles, because it is bounded above by a function of the two members’ error rates: 10 of our 12 within-axis pairs sit exactly at that bound. Looking at the error sets directly explains why. In 44 of 66 configuration pairs, one member’s errors are a strict *subset* of the other’s, so the weaker member recovers nothing the stronger one missed and a majority vote has no material to work with. That nesting holds for 27 of the 28 pairs drawn from a single base model and falls to about half for pairs spanning different labs, against a permutation baseline below 1% ($p < 0.0005$). The consequence is measurable at the level practitioners care about: majority voting fails to beat the single best member of its own ensemble on any axis, trailing it by up to 20 points. We recommend reporting the fraction of ensemble member pairs whose errors are nested alongside any correlation-based diversity claim, since it is a direct count that cannot be inflated by adding a weaker member.

Index Terms—large language models, ensemble diversity, self-consistency, LLM-as-judge, multi-agent debate, effective sample size, reliability

I. INTRODUCTION

A. Background and Motivation

Self-consistency decoding, multi-agent debate, and LLM-as-judge ensembles all treat each sampled response as an independent vote. In practice, many of these pipelines draw every sample from the same underlying base model, varying only a shallow surface property such as the system prompt or the sampling temperature. If those samples share correlated failure modes, the effective diversity of the ensemble is far smaller than its nominal size N suggests. Majority voting, confidence aggregation, and judge panels then inherit a reliability ceiling that is invisible to the people building them.

B. Challenges in Quantifying Ensemble Diversity

Diversity is easy to induce along superficial axes (a different persona in the system prompt, a higher decoding temperature) and hard to induce along axes that plausibly decorrelate errors (a genuinely different training run, a different base model family). Prior evaluation practice rarely distinguishes between these axes’ contribution to ensemble reliability, and error correlation across ensemble members is rarely reported at all, leaving practitioners without a way to tell whether adding a 4th or 8th sample from the same model is buying real reliability or merely restating the 1st sample with different words.

C. Role of N_{eff} in This Work

We adapt the effective-sample-size framework used for correlated estimators to LLM ensembles. The quantity is Kish’s design effect [7], standard in survey sampling for discounting a nominal sample size by the correlation among its units: given N ensemble members with mean pairwise error correlation $\bar{\rho}$,

$$N_{\text{eff}} = \frac{N}{1 + (N - 1)\bar{\rho}}. \quad (1)$$

N_{eff} answers a concrete question: how many *independent* voters does this ensemble behave like? It also lets us compare diversity axes on a common scale rather than only on raw accuracy. We compute N_{eff} for 12 configurations spanning 4 axes (3 configurations per axis: prompt-diverse, temperature-diverse, size-diverse, and family-diverse), with bootstrap confidence intervals over questions, and report where each axis’s N_{eff} saturates as N grows.

D. Contributions

- We decompose measured ensemble independence by *construction axis*, varying one practitioner-controllable choice at a time rather than auditing diversity in aggregate.
- We show the statistic normally used is biased in a specific, correctable direction: ϕ is bounded above by a function of the members’ error rates (Eq. 2), so members differing in accuracy are credited with independence they do not have. 10 of our 12 within-axis pairs sit exactly at that bound.

- We identify the structure responsible: errors are largely *nested* rather than overlapping. In 44 of 66 pairs one member’s errors are a strict subset of the other’s, including 27 of the 28 same-base-model pairs, against a permutation baseline below 1% ($p < 0.0005$). A nested pair has zero complementary errors, so no aggregation rule over it can recover a question.
- We show the nesting rate is far higher within a single base model than across models (96% against 41% and 48% for pairs sharing a lab and spanning labs; trend $p = 1.7 \times 10^{-4}$), which gives the practitioner guidance a correlation ranking only gestures at.
- We connect the structure to the outcome it predicts. Majority voting never beats the best member of its own ensemble on any axis, trailing it by up to 20 points, and outside family-diverse only 1 to 2 questions per ensemble were recoverable by a better aggregation rule.
- We compare five allocation strategies at equal API cost under all three measures, and show that the ranking a correlation-based N_{eff} would suggest does not survive the correction (a 32% apparent advantage for cross-family ensembling becomes about 2%). We release the harness, the graded error matrix, and a decision log recording every deviation from the frozen specification.

II. RELATED WORK

Self-consistency decoding [1] samples multiple chain-of-thought [2] completions from a single model and takes a majority vote, implicitly treating the samples as independent draws. Multi-agent debate frameworks [3] extend this to multiple model instances arguing over several rounds, again typically instantiated with a single underlying model. LLM-as-judge evaluation [4] raises the same concern from the evaluation side: a judge panel built from one base model may share the judge’s own biases across every panelist. A parallel line builds explicit LLM ensembles for accuracy: Agent Forest [5] scales by sampling and voting over instances of one model, and Mixture-of-Agents [6] layers multiple LLMs. Neither measures the pairwise error correlation its aggregation step relies on, so a reported gain from more samples is not distinguishable, from the published results alone, from the diminishing-returns pattern we measure in Sec. IV.

Classical ensemble theory holds that accuracy gains are governed by the trade-off between member accuracy and pairwise error correlation: the ambiguity decomposition [8] formalized it, and Dietterich [9] and Kuncheva and Whitaker [10] survey the constructions that induce decorrelated errors and the diversity measures used to quantify them. We apply the same design-effect statistic to a setting that theory did not anticipate: API-only, black-box ensembles whose members cannot be inspected, retrained or explicitly decorrelated, only reconfigured along the surface axes an API caller is given.

Two concurrent 2026 studies confirm the independence assumption is unreliable, from different angles. Kim [11] audits five diversity metrics across 31,900 ensemble subsets on MMLU-Pro and finds that apparent diversity gains are largely

capability entanglement rather than genuine error decorrelation. Once capability is controlled for, simple majority voting beats the best individual model in under 10% of size-3 subsets. Ding [12] audits cross-model agreement as a confidence signal at large scale (265,000 samples on GPQA-Diamond and AIME) and reports only weak correlation (0.20–0.59) between agreement and correctness, concluding that self-consistency is a conditional proxy rather than a standalone confidence score. Both results corroborate our starting premise, that ensemble diversity is routinely overestimated, but neither decomposes *why*: they do not attribute the correlation they measure to specific, practitioner-controllable construction choices. Our contribution is narrower: we hold N_{eff} fixed and vary one structural axis at a time to localize *where* an ensemble builder’s choices buy real independence. We also sharpen Kim’s [11] conclusion. That work attributes apparent diversity gains to capability entanglement, established by controlling for capability across ensemble subsets. Sec. IV-B identifies a mechanism operating even within a fixed subset: the correlation coefficient is itself bounded by the members’ error rates, so two members of unequal accuracy *cannot* record high correlation however similarly they fail. The entanglement is therefore a property of the measurement, not only of the sampling, and it is correctable.

III. METHODOLOGY

A. Datasets

We evaluate on three public benchmarks spanning multiple-choice and free-response formats: GPQA-Diamond [13], a graduate-level multiple-choice science benchmark; MATH [14] restricted to levels 4–5 (the hardest tier); and SimpleQA [15], a short-form factual QA benchmark. Free-tier API quotas (Sec. V) constrained the question set to $N = 30$ total (8 GPQA-Diamond, 11 MATH levels 4–5, 11 SimpleQA), a deliberate reduction from an original target of 800 questions. See `logs/decisions.md` for the two intermediate reductions (800→150→30) and the quota measurements that drove each.

B. Diversity Axes and Configurations

We define 12 configurations across 4 diversity axes, 3 configurations per axis, per the frozen specification in `config/project_spec.yaml`:

- **Prompt-diverse**: one fixed model at temperature 0.7, with three system-prompt personas (“careful analyst”, “confident expert”, “skeptical reviewer”).
- **Temperature-diverse**: the same fixed model and a neutral prompt, at temperatures $\{0.3, 0.7, 1.0\}$.
- **Size-diverse**: one model family, 3 sizes (small, medium, flagship), neutral prompt, temperature 0.7.
- **Family-diverse**: 3 different model providers/labs, neutral prompt, temperature 0.7.

Each (question, configuration) pair receives exactly one sample, since the goal is to measure cross-configuration error correlation rather than within-configuration sampling variance.

One consequence is worth stating plainly: for the temperature axis, a single draw per configuration cannot separate temperature-induced diversity from ordinary sampling noise at that temperature, so that axis measures the diversity a practitioner actually gets from one sample per setting rather than the full sampling distribution.

C. Grading

Each dataset is graded by its own deterministic procedure. GPQA-Diamond uses exact match on the extracted answer letter. MATH uses symbolic equivalence via `math_verify` on the extracted `\boxed{}` expression, so that algebraically equal forms are not counted as disagreements. SimpleQA uses normalized string containment against the reference answer, with an LLM judge as a fallback only when containment fails.

That fallback deserves an explicit caveat, because the judge is `gemini-3.5-flash`, which is also the base model behind 8 of the 12 configurations. A model therefore participates in grading responses that include its own. We mitigate this three ways. The judge is a fallback rather than the primary path, every invocation is written to an audit log released with the paper, and, as a robustness check, we recompute the paper’s main result with SimpleQA excluded entirely (Sec. IV-C). We would not use a self-judging setup by choice; it is a consequence of the free-tier constraint that no second model with sufficient daily quota was available.

D. Effective Independence Metric

For any subset of configurations we compute the pairwise error-correlation matrix and N_{eff} via Eq. 1, with 95% confidence intervals from 1,000 bootstrap resamples over questions. We additionally report the fraction of questions on which $\geq 80\%$ of an ensemble agrees and is wrong. This confident-correlated-error rate is the failure mode majority voting cannot detect or correct. Finally, we compare a fixed set of five allocation strategies at a matched budget of $k = 3$ ensemble members (Sec. IV-E), which re-uses the cached results already generated for the axis comparison and issues no additional API calls.

E. Experimental Harness

The harness, the frozen specification, the model configuration, the graded error matrix, and the full decision log are released with this paper at <https://github.com/reinamercy/Effective-Independence>. File paths given below refer to that repository. The decision log in particular records every deviation from the original specification, in the order the deviations happened and with the reasoning at the time.

All generations are produced through a cached, rate-limited API client (one persisted response per unique model/prompt/temperature/seed tuple), so no (question, configuration) pair is ever re-queried. Every API call is logged with the provider-returned model version string, token counts, and cost. All generation uses the free-tier model set frozen in `config/models.yaml`. The prompt- and temperature-diverse axes both use `gemini-3.5-flash` [16], which is

also the medium tier of size-diverse; the small and flagship tiers of size-diverse are `gemini-3.5-flash-lite` and `gemini-3.7-flash` [16]. Family-diverse combines `gemini-3.5-flash` [16], `z-ai/glm-5.2` [17], and `nvidia/nemotron-3-ultra-550b-a55b` [18]. (The OpenAI slot originally used `openai/gpt-oss-20b` but was replaced with `z-ai/glm-5.2` mid-project after OpenRouter permanently de-listed the former from its free tier; see the decision log.) Dataset loading, configuration construction, cached generation, per-dataset grading and N_{eff} analysis are independently rerunnable modules connected only through on-disk artifacts, so any stage can be re-run in isolation.

IV. RESULTS

A. Quantitative Analysis

All results are computed on the complete 30×12 design: all 360 (question, configuration) cells were generated and graded, at 75.0% overall accuracy. Every pairwise comparison below therefore rests on the same 30 questions, with no missing cells and no coverage confound. Table I reports N_{eff} per axis with bootstrap 95% confidence intervals. Prompt-diverse and temperature-diverse, the two axes that vary only a surface property of one fixed base model, show the highest mean pairwise error correlation ($\bar{\rho} = 0.88$ and 0.87) and collapse to $N_{\text{eff}} \approx 1.1$ regardless of their nominal size $N = 3$. Size-diverse is somewhat better ($\bar{\rho} = 0.71$, $N_{\text{eff}} = 1.24$), consistent with `config/models.yaml`’s documented caveat that its “flagship” tier is a newer Flash-generation model rather than a genuinely larger one, so this axis likely understates what true parameter-scale diversity would buy. Family-diverse scores best: mean error correlation drops to $\bar{\rho} = 0.55$ and N_{eff} rises to 1.43, still far from the nominal $N = 3$. Extending this to the full 12-configuration ensemble, N_{eff} reaches only 1.37: adding nine further configurations beyond $N = 3$ buys roughly a third of one additional effective voter.

TABLE I
 N_{EFF} PER AXIS UNDER ϕ (BOOTSTRAP 95% CI) AND UNDER THE MARGINAL-ROBUST $\bar{Q}$, ON THE COMPLETE 360-CELL DESIGN. EVERY PAIR IS MEASURED ON ALL 30 QUESTIONS. THE ϕ COLUMNS ARE INFLATED BY UNEQUAL MEMBER ACCURACY (SEC. IV-B).

Axis	$\bar{\rho}$	$N_{\text{eff}}(\phi)$	95% CI	$\bar{Q}$	$N_{\text{eff}}(Q)$
Prompt-diverse	0.88	1.08	[1.00, 1.27]	1.00	1.00
Temperature-div.	0.87	1.10	[1.00, 1.29]	1.00	1.00
Size-diverse	0.71	1.24	[1.07, 1.58]	0.99	1.00
Family-diverse	0.55	1.43	[1.23, 1.75]	0.98	1.02
All 12 configs	0.71	1.37	[1.18, 1.82]	0.97	1.03

We also measure the confident-correlated-error rate: the fraction of questions where $\geq 80\%$ of a ≥ 3 -member ensemble is simultaneously wrong. Across all 12 configurations this occurs on 5 of 30 questions (16.7%), and per axis ranges from 13.3% (size-diverse) to 20.0% (prompt-diverse). A majority vote over any of these ensembles returns the wrong answer confidently on those questions; diversity in surface presentation does not protect against it. Per-configuration accuracy

varies almost as much within an axis as across axes (individual configurations span 53.3% to 83.3%), which matters because an axis can spread its members widely on accuracy and still have highly correlated errors: N_{eff} responds to *which* questions each member gets wrong, not how many.

B. Correlation Is Capped by Accuracy Differences

A second constraint limits what the correlation numbers can support, and it cuts deeper than sample size. The phi coefficient between two binary error vectors is bounded by their marginals. If configuration A is wrong on a fraction p of questions and B on a fraction q , with $p \leq q$, then even when A 's errors are a strict subset of B 's, which is the most concordant arrangement available,

$$\phi_{\max} = \sqrt{\frac{p(1-q)}{q(1-p)}}, \quad (2)$$

which equals 1 only when $p = q$. The bound follows from maximising $\phi = (n_{11}n_{00} - n_{10}n_{01})/\sqrt{pq(1-p)(1-q)}$ over the 2×2 tables compatible with fixed marginals: the numerator is largest when the smaller error set is contained in the larger, which drives one off-diagonal cell to zero and leaves Eq. 2. Two configurations with different accuracy therefore *cannot* record a high correlation, however similarly they fail. A low ϕ is evidence of genuine error decorrelation only when it sits appreciably below this ceiling.

Checking each observed correlation against its own ceiling changes how Table I should be read. Ten of the twelve within-axis pairs sit exactly at their marginal-bound ceiling: their errors are already as correlated as their differing accuracy permits. This includes two of the three family-diverse pairs, where one member is markedly more accurate than the other. For those two, Eq. 2 caps ϕ at 0.511 and 0.478, and those are precisely the observed values. Much of the low correlation on this axis is therefore explained by its members having unequal accuracy rather than by their failing on complementary questions.

Two of the twelve within-axis pairs fall clearly below their ceiling, the clearest being the two open-weight models at $\phi = 0.665$ against a bound of 0.935. Among the within-axis pairs these are the only evidence of error decorrelation not attributable to an accuracy mismatch. (Sec. IV-C widens the view to all 66 pairs, where cross-lab pairs supply further such cases.) We therefore do not claim that cross-family ensembling decorrelates errors. The defensible reading is narrower: measured correlation is close to the maximum the data permits for most pairs, and much of what separates the axes in Table I is how unequal their members' accuracies are.

This does not undermine the N_{eff} values themselves, which describe the actual correlation structure whatever produces it, but it does undermine any causal reading of the axis ranking. It is also a caution: an ensemble can post a lower $\bar{\rho}$, and so a higher N_{eff} , simply by including a weaker member.

C. Errors Are Nested, Not Complementary

The bound in Eq. 2 is tight precisely when one member's errors are a subset of the other's, so the finding that most pairs sit at their ceiling is really a claim about the *structure* of the error sets. That structure can be counted directly, without estimating a coefficient at all, and doing so turns out to be both more robust at this sample size and more informative about what an ensemble can do.

Call a pair *nested* when one member's error set contains the other's, i.e. one of the two off-diagonal cells of the 2×2 error table is empty. Pairs in which either member has no errors at all over the shared questions are excluded rather than counted as nested, since such a member is trivially contained in anything and would inflate the rate. Nesting is the case that matters operationally: if A 's mistakes are a subset of B 's, then B never answers a question correctly that A gets wrong, so no aggregation rule over $\{A, B\}$ can recover anything A missed; Sec. IV-D tests that consequence directly against majority voting. The pair is ordered by capability rather than diversified by it. A pair with complementary errors on both sides, by contrast, gives a vote something to work with.

The result is stark. In 44 of 66 configuration pairs the errors are fully nested (bootstrap 95% CI over questions [67%, 95%]). Table II breaks this down by how genuinely different the two members are: **27 of the 28 pairs drawn from the same base model are nested**, 96% (Wilson 95% CI [82%, 99%]), against 41% for pairs sharing a lab but not a model ([22%, 64%]) and 48% for pairs spanning different labs ([28%, 68%]). A Cochran–Armitage test across the three ordered groups confirms a decreasing trend ($z = -3.77$, $p = 1.7 \times 10^{-4}$), though the decrease is not strictly monotonic: the two cross-model categories are statistically indistinguishable from one another, and the separation that drives the trend is between same-model pairs and everything else. Fig. 1 plots the rates.

One scoping caveat belongs here, not only in Limitations. Eight of our twelve configurations run on `gemini-3.5-flash`, so the same-model row characterises one model observed under eight configurations rather than a property established across many base models. Whether it generalises is an open question our model set cannot answer.

Sparse tables could produce empty off-diagonal cells by luck, so this rate means nothing without a baseline. We therefore permute, holding each configuration's error count and its set of graded questions fixed and shuffling only *which* of those questions it fails. That destroys any real shared failure mode while preserving the marginals and the coverage pattern exactly. Over 2,000 permutations the expected nesting rate is under 1% in both cases (95% range [0%, 2%] across all pairs and [0%, 0%] among same-model pairs). Against observed rates of 67% and 96%, no permutation in either null came close to the observed value ($p < 0.0005$). At full coverage the chance baseline is far lower than it was under partial coverage, because empty off-diagonal cells stop being easy to produce by luck once every pair is measured on all 30 questions. The nesting is a property of the models, not of the sample size.

A second control addresses the grading circularity of Sec. III-C. Excluding SimpleQA entirely leaves 19 questions, where nesting rises to 91% (60 of 66 pairs) and same-model pairs remain at 27 of 28. The effect is not produced by the self-judged dataset; if anything it is stronger without it.

The first row is the paper’s sharpest result. Across 27 of the 28 pairs sharing a base model, there is not one question the second configuration gets right that the first gets wrong. Changing persona or temperature reorders and rewords the output, and changes how often the model is wrong, but in all but one pair it never contributes an independent success. That is a stronger and more directly checkable claim than $\bar{\rho} = 0.88$, since it is a count over the complete design rather than an estimated coefficient.

It also explains the earlier numbers. Nested pairs are exactly those pinned to the Eq. 2 ceiling, so the N_{eff} spread in Table I largely tracks how unequal the members’ accuracies are while the underlying dependence is close to total throughout. Recomputing with Yule’s Q , which is not marginal-bounded and reaches 1 whenever an off-diagonal cell is empty, makes this explicit: every axis falls to $N_{\text{eff}} \approx 1.0$ (Table II), the 12-configuration ensemble to 1.04.

We are explicit about that substitution’s status. Eq. 1 is derived for a pairwise *correlation*, and Q sits on a different scale, so $N_{\text{eff}}(Q)$ is a diagnostic rather than a drop-in estimator: it shows that the ordering and magnitudes implied by ϕ do not survive a measure free of the marginal cap. Q also saturates at exactly 1 whenever an off-diagonal cell is empty, so with as few as five errors in some pairs a single flipped answer would move it. The nesting counts, which involve no formula, therefore carry the claim; $N_{\text{eff}}(Q)$ expresses it on a familiar scale. Both point the same way, and further than ϕ does.

TABLE II
ERROR NESTING ON THE COMPLETE 360-CELL DESIGN. A PAIR IS NESTED WHEN ONE MEMBER’S ERRORS ARE A SUBSET OF THE OTHER’S, SO THE PAIR HAS NO COMPLEMENTARY ERRORS FOR A VOTE TO EXPLOIT.

Pair type	Nested pairs	Rate
Same exact model	27 / 28	96%
Same lab, different model	7 / 17	41%
Different lab	10 / 21	48%
All pairs	44 / 66	67%

D. Does Majority Voting Actually Help?

Everything above characterises the ensemble through a statistic or a structure. Neither is what a practitioner buys. The question they face is whether aggregating k samples answers more questions correctly than using one model, so we measure that directly. It requires no additional generation: majority vote is computed from the same graded cells, with an even split counted as incorrect. To keep the panel constant across questions, each ensemble is scored only on questions where all of its members are graded.

Table III reports the result, and it is one-sided. **Majority voting does not beat the single best member of its own**

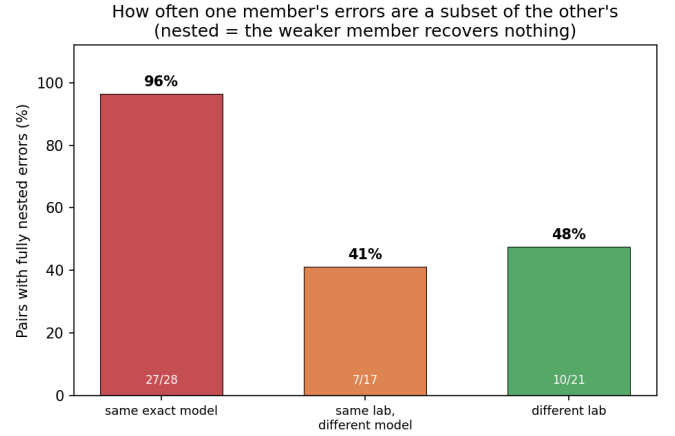


Fig. 1. How often one member’s errors are a strict subset of the other’s, by how genuinely different the two members are on the complete 360-cell design.

ensemble on any axis. It loses by 3.3 points on prompt-, temperature- and size-diverse alike, and by 20.0 points on family-diverse. Over all 12 configurations together it reaches 80.0% against 83.3% for the best member, a deficit of 3.3 points at twelve times the API cost.

This is the outcome the nesting structure predicts. When one member’s errors are a subset of another’s, the ensemble is ordered by capability, and a vote can only dilute the strongest member with weaker ones that contribute no question it missed. The recoverable column shows where the exception lies: for the three same-base-model axes only 1 or 2 questions were available for a better aggregation rule to rescue, so their failure is not a deficiency of majority voting specifically but an absence of complementary errors to aggregate. Family-diverse is the one ensemble with real material to work with, 6 recoverable questions, and it is also the one where voting loses by the widest margin: its members disagree often enough to outvote the strongest of them.

The obvious objection is that the best member is identifiable only in hindsight; a practitioner choosing in advance is more fairly compared against the *mean* member, which voting matches or beats everywhere except family-diverse. The honest summary is narrower than “voting is useless”: voting recovers roughly the average member’s accuracy, never an accuracy no individual member already had.

E. Budget-Matched Allocation

The axis comparison above asks which kind of diversity decorrelates errors. The question a practitioner actually faces is narrower: given a fixed number of API calls per question, how should they be spent? We therefore compare five allocation strategies at a matched budget of $k = 3$ calls. Every strategy issues exactly the same number of requests, so any difference in N_{eff} is attributable purely to how the budget was allocated rather than to how large it was.

We note explicitly that the frozen specification framed this comparison as one at matched total API *cost*. Every model

TABLE III

MAJORITY VOTE AGAINST THE MEMBERS IT AGGREGATES, ON THE COMPLETE 360-CELL DESIGN. “RECOV.” COUNTS QUESTIONS WHERE THE MAJORITY WAS WRONG BUT AT LEAST ONE MEMBER WAS RIGHT. A NON-POSITIVE VOTE—BEST COLUMN MEANS VOTING NEVER BEAT THE STRONGEST MEMBER.

Ensemble	n_q	Vote	Best	Mean	Recov.
Prompt-diverse	30	76.7	80.0	76.7	1
Temperature-diverse	30	80.0	83.3	80.0	1
Size-diverse	30	80.0	83.3	78.9	2
Family-diverse	30	63.3	83.3	64.4	6
All 12 configs	30	80.0	83.3	75.0	2

in the final configuration is free-tier, so dollar cost cannot be the matching variable; we match on k instead, which is both the genuinely scarce resource in this project (per-model daily request quotas) and the quantity a practitioner controls. This substitution is recorded in `logs/decisions.md` rather than made silently.

Table IV reports each allocation under all three measures. Reporting only the ϕ column here would contradict Sec. IV-B, and the disagreement between the columns is large enough that it changes the conclusion.

Read through ϕ , cross-family looks decisively best: $N_{\text{eff}} = 1.43$ against 1.08 for three prompts of one model, a 32% gain at identical cost. Almost all of that gap is the accuracy artifact. Under Q the same comparison is 1.02 against 1.00. We therefore do not claim a 32% reliability gain from crossing model families, and a practitioner should not budget for one.

What survives the correction is visible in the last column. The two allocations drawn from a single base model are fully nested, 3 pairs of 3: no member ever answers a question its partners miss. Every allocation that spans more than one model contains at least one pair with complementary errors. That is the real distinction at a fixed budget, and it is a difference in kind rather than a 32% improvement in a statistic.

This also corrects a reading ϕ invites. Under ϕ the mixed allocation (1.18) looks much worse than cross-family (1.43), suggesting that spreading a budget across axes poorly substitutes for changing model family. Under Q the two are close (1.01 and 1.02). What matters is whether the allocation spans more than one base model at all, rather than how many *kinds* of diversity it mixes. With differences this small at $N = 30$, we do not rank the non-nested allocations against each other.

V. LIMITATIONS

Model set. Budget constraints limited us to free-tier providers (Sec. III-E) rather than the closed-frontier set originally planned, after we found the real free-tier quota (20 requests/day/model) far below what public documentation reports. Family-diverse therefore mixes closed-weight (Gemini) and open-weight (GLM, Nemotron) models rather than three frontier flagships, and its members are the most unequal in accuracy of any axis, which interacts with the bound below. **Dataset size.** With coverage complete, the binding limitation is the question set: $N = 30$ questions (~ 8 –11 per dataset)

TABLE IV

ALLOCATIONS AT A MATCHED BUDGET OF $k = 3$ API CALLS PER QUESTION, ON THE COMPLETE 360-CELL DESIGN. EVERY ROW COSTS THE SAME. THE ϕ COLUMN OVERSTATES THE SPREAD BECAUSE IT IS CAPPED BY DIFFERENCES IN MEMBER ACCURACY (SEC. IV-B); THE Q AND NESTING COLUMNS ARE THE ONES TO READ.

Allocation ($k = 3$)	$N_{\text{eff}}(\phi)$	$N_{\text{eff}}(Q)$	Nested
Same model, 3 prompts	1.08	1.00	3/3
Same model, 3 temperatures	1.10	1.00	3/3
Same family, 3 sizes	1.24	1.00	2/3
Mixed, one per axis	1.18	1.01	1/3
3 different families	1.43	1.02	2/3

leaves several 95% intervals wide (size-diverse: $[1.07, 1.58]$). This is a power limitation rather than an artifact of the method, so the axis *ordering* should be read as descriptive of this dataset rather than as a population ordering. A larger question set is the single most valuable extension. **Both association measures are constrained.** ϕ cannot exceed the Eq. 2 ceiling and 10 of 12 within-axis pairs sit exactly there, so differences in $\bar{\rho}$ largely track how unequal member accuracies are; a design holding accuracy constant would separate the effects, but the free-tier set fixes which models exist at each tier. Q conversely saturates at 1 whenever an off-diagonal cell is empty, and with as few as five errors in some pairs one flipped answer would move it. The nesting counts are immune to both problems. **Nesting is pairwise.** Three configurations can be pairwise nested in a chain with no single member dominating all others; our counts do not distinguish that case, and the operational claim we make is the pairwise one. **Single sample per cell.** We measure cross-configuration correlation, not within-configuration sampling variance, so for the temperature axis in particular one draw per setting cannot separate temperature-induced diversity from sampling noise. **A grading judge shares a base model with the configurations.** SimpleQA falls back to an LLM judge when string containment fails, and that judge is `gemini-3.5-flash`, which underlies 8 of the 12 configurations (Sec. III-C). Excluding SimpleQA leaves the nesting result intact (91% overall, same-model pairs still 27 of 28), so we do not believe it drives the finding, but an independent grader would be preferable and was unavailable within the quota.

VI. CONCLUSION

On the complete design, prompt- and temperature-diversity are the cheapest and most commonly used axes, and the two that buy the least real independence: $N_{\text{eff}} \approx 1.1$ at nominal $N = 3$. Model-family diversity is the axis on which measured correlation falls furthest, but we do not claim the gain its ϕ -based N_{eff} of 1.43 implies: that margin rests largely on pairs whose low correlation is already explained by unequal member accuracy (Sec. IV-B). Corrected, the comparison is 1.02 against 1.00.

The ranking is the least robust part of these results. The sturdiest finding is structural and needs no coefficient: in 44 of 66 pairs one member’s errors are a strict subset of the other’s,

including 27 of the 28 pairs drawn from the same base model. Where errors are nested there is nothing to recover, which is why a nominal three or twelve votes behaves like roughly one. The prediction is borne out directly: majority voting never beat the best member of its own ensemble on any axis, and outside family-diverse only 1 to 2 questions per ensemble were even available for a smarter aggregator to rescue.

For debate, self-consistency, and judge-panel pipeline design, that means scaling the number of same-base-model samples is a much weaker reliability lever than its nominal N suggests. At a fixed budget the distinction our data supports is categorical rather than a matter of degree: allocations drawn entirely from one base model are fully nested and can recover nothing, while allocations spanning more than one base model contain at least one pair with complementary errors. How many different *kinds* of diversity are mixed matters less than whether the base model changes at all.

Measuring that structure carries its own trap: because ϕ is capped by the members’ error rates, an ensemble can report a flattering N_{eff} merely by including a weaker member. The remedy needs no extra API calls. Alongside any reported $\bar{\rho}$ or N_{eff} , publish the fraction of member pairs whose errors are nested: a direct count, uninflatable by adding a weak member, answering what a builder needs to know.

These remain small-scale results at $N = 30$ questions (Sec. V), consistent in direction with classical ensemble theory and with concurrent independent work [11], [12]. The design is now complete, so the numbers no longer move with coverage, but a larger question set would be needed before treating the axis ordering as settled.

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